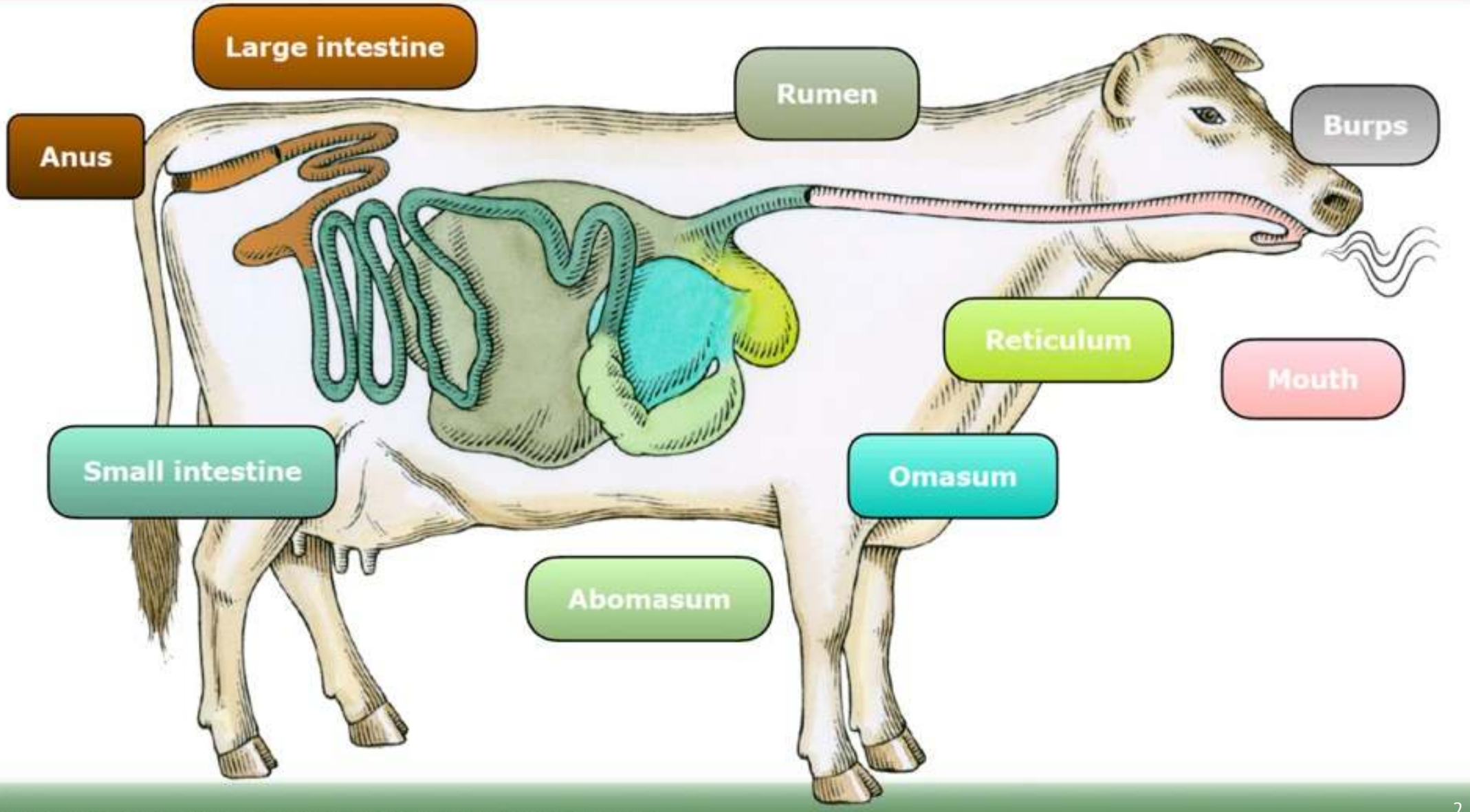
Ruminant Disease

Clinical Exanimation in ruminants

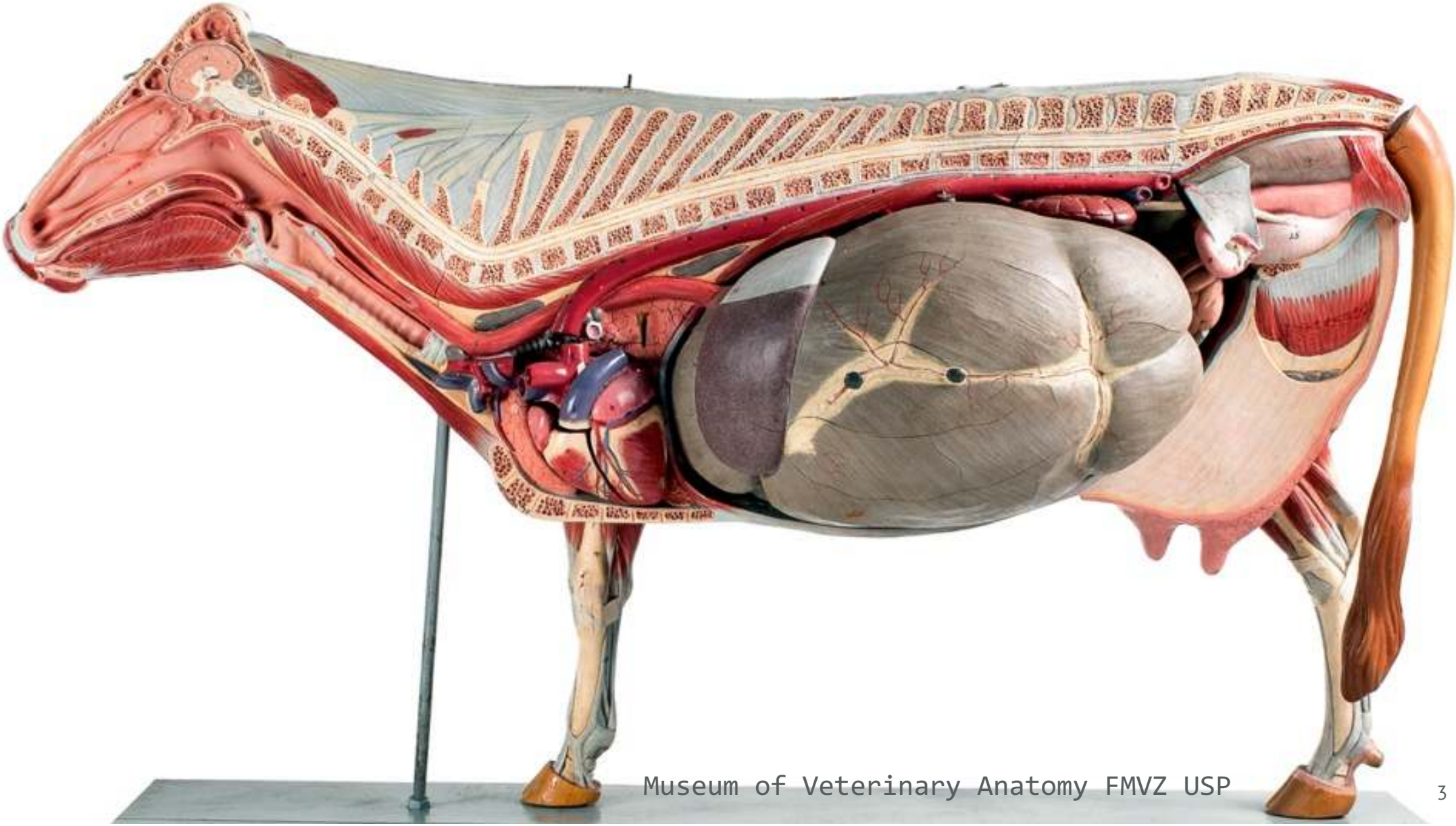


The special stomach in ruminants

Ruminants (like cattle, sheep, goats, and deer) have one large stomach divided into four highly specialized compartments: the rumen, the reticulum, the omasum (foregut, 前胃), and the abomasum. This complex digestive setup allows them to ferment and extract nutrients from tough, fibrous plant materials (like grass and hay) that simple-stomached animals cannot break down.



© DairyNZ and The University of Waikato Te Whare Wānanga o Waikato



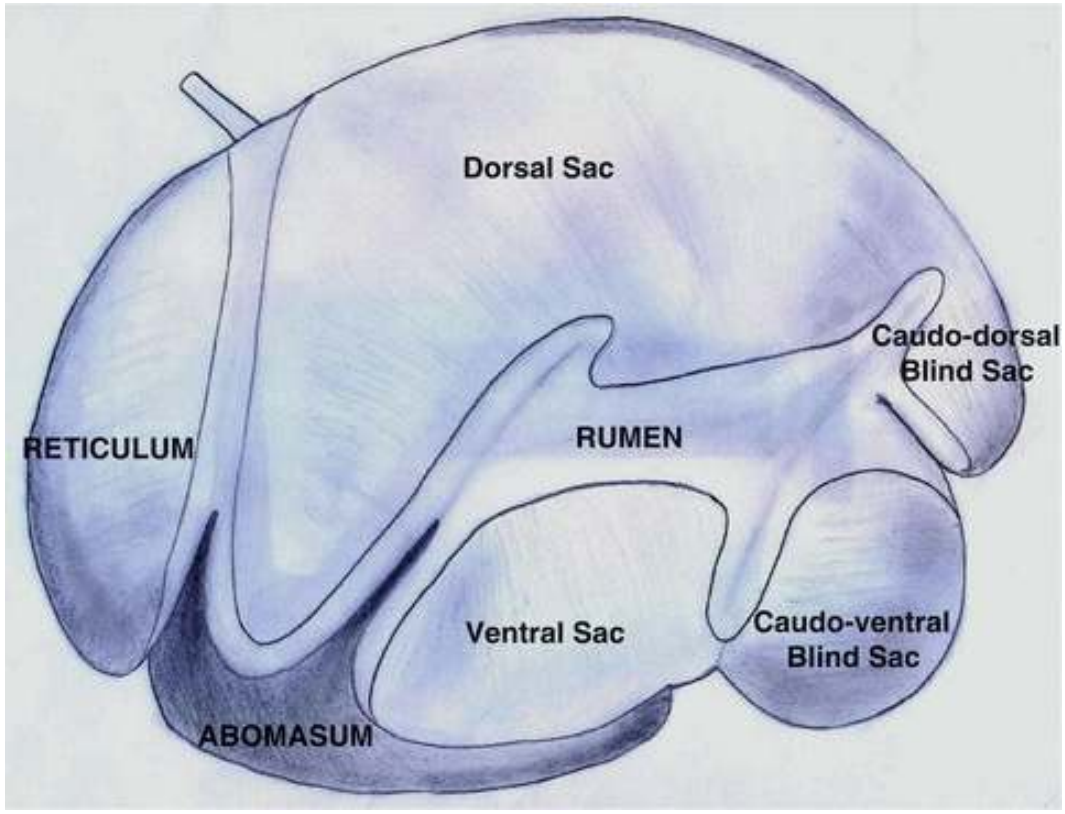
Museum of Veterinary Anatomy FMVZ USP



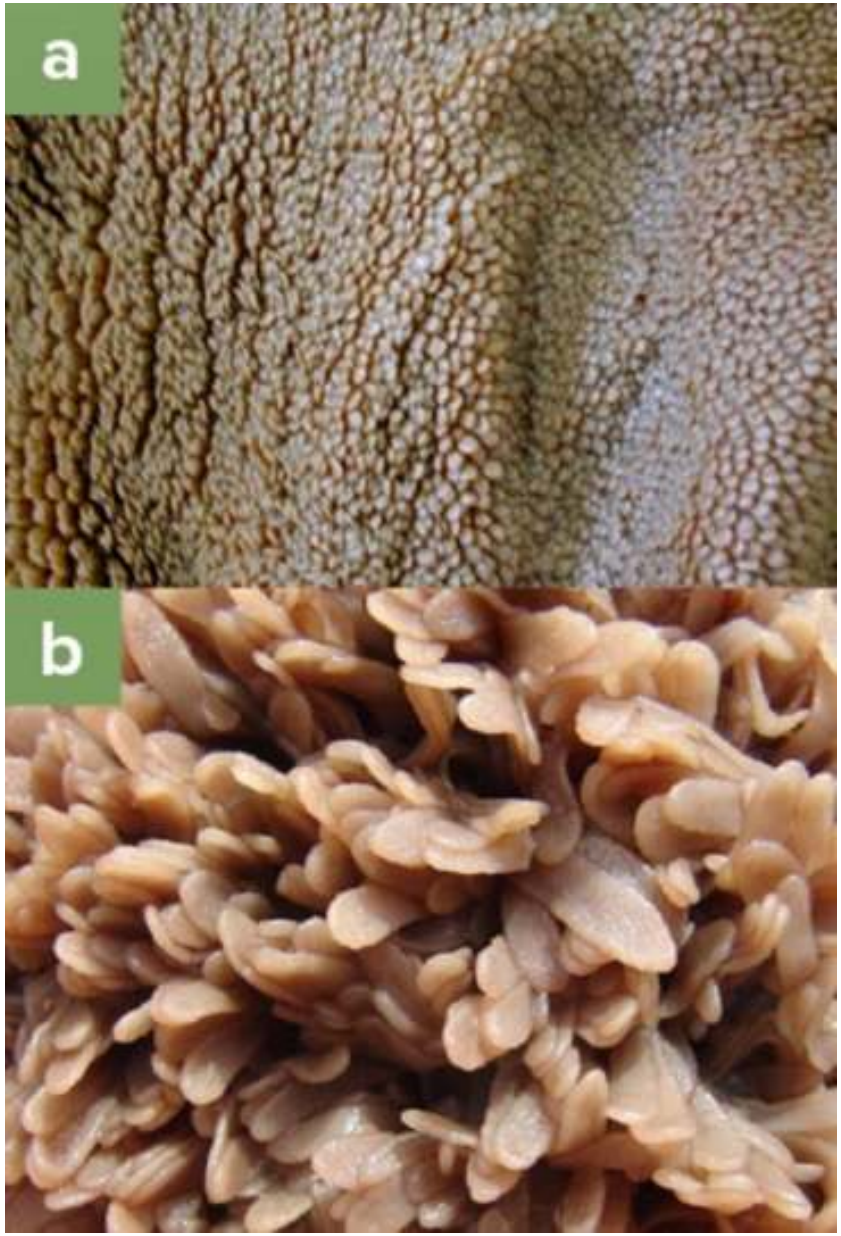


Rumen (瘤胃)

- Main digestive organs of ruminants, 90–120 L
- Countless symbiotic microorganisms
  - digest feed into VFAs
- Gas, liquid, solid (fiber mat) phase



https://doi.org/10.1007/978-3-319-30533-2_1



(adopted from Dr. Petersen K, Dr. Alonso L, and Dr. Malafaia P)

Rumen

- Ideal pH value: 6.5–6.8.
- Rumination promotes saliva secretion
  - pH of 8.0–8.5: contained carbonates and phosphates, as buffer for the rumen environment
  - at least 130–150 L (and even up to 220–250 L) per day (Maekawa *et al.*, 2002).



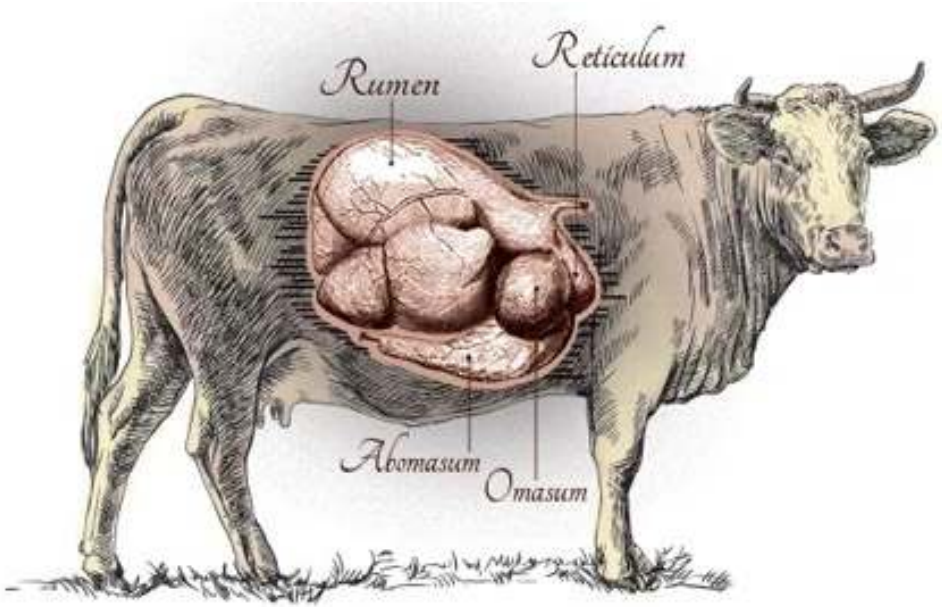
Rumination (反芻)

- Time and frequency: depend on the crude fiber percentage and length and the number of feedings
- Rumination takes at least 7 hours a day.

Reticulum (蜂巢胃)

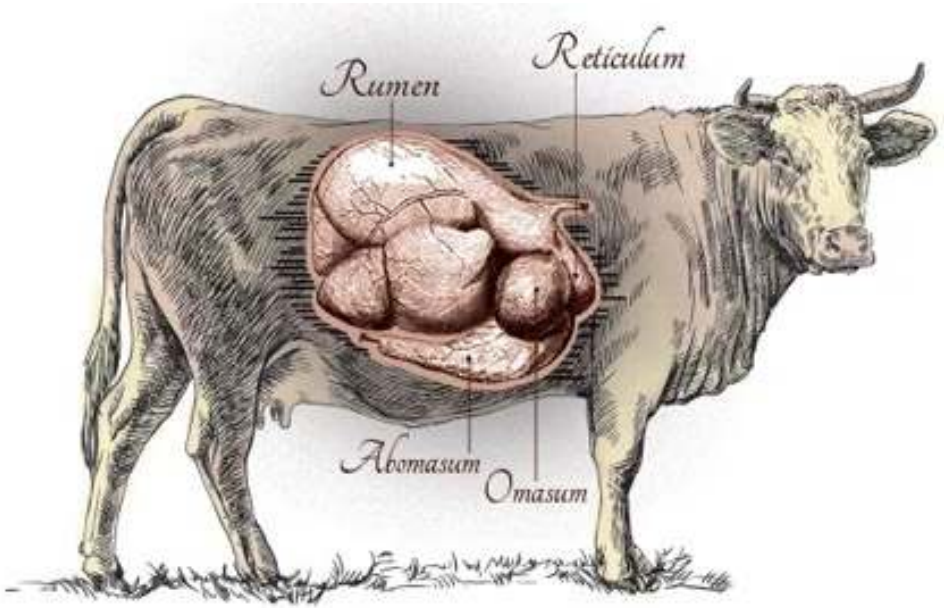


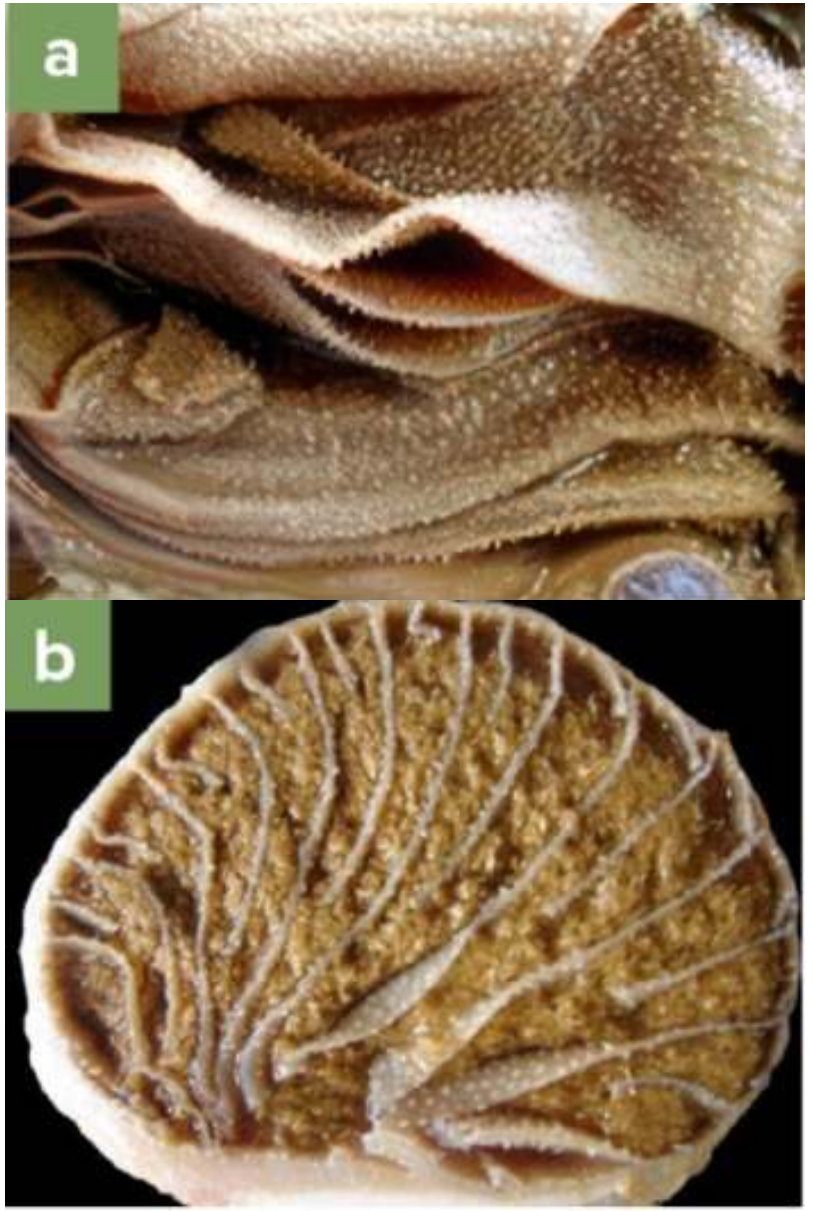
(adopted from Dr. Petersen K, Dr. Alonso L, and Dr. Malafaia P)





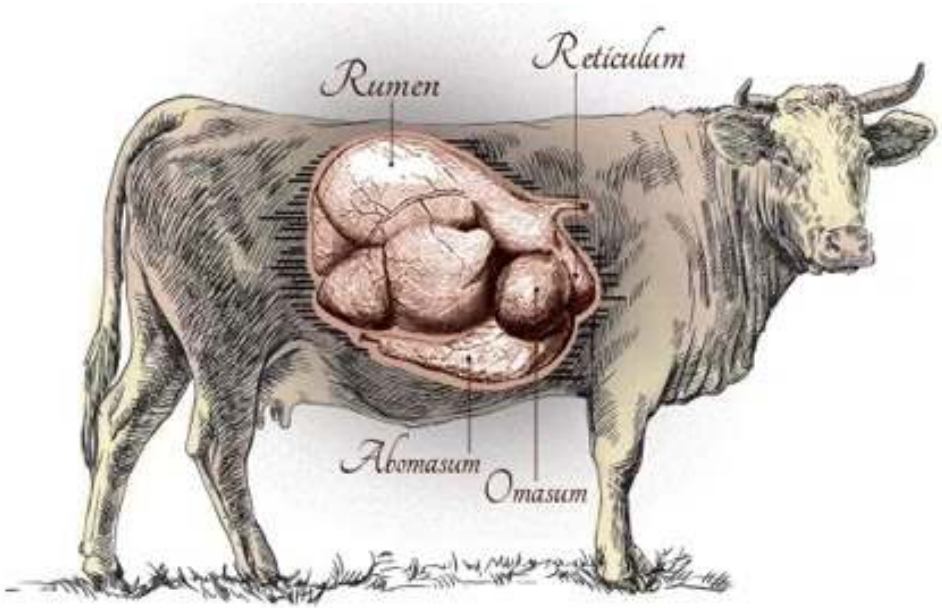
Omasum (重辦胃)





(adopted from Dr. Petersen K, Dr. Alonso L, and Dr. Malafaia P)

Abomasum (皺胃)





(adopted from Dr. Petersen K, Dr. Alonso L, and Dr. Malafaia P)

Clinical Examination_Obtain the History

- The information of the chief or presenting complaint
  - duration, gradual or sudden
  - last parturition date (female)
  - production parameters (for dairy animals)
  - other associated signs
  - what and how the animal fed (refuse any or all of the feed)

Clinical Examination_Obtain the History

- The information of the chief or presenting complaint
  - treatment history
  - previous diseases noted in the herd
  - vaccination/worming history
  - pasture or housing practices

Clinical Examination

Physical examination provides the veterinarian with information that is used to assess the health status of the patient. The information combined with the history enables us to determine what the problems are and help to focus the disease on specific organ systems.

Clinical Examination

In some cases, such as economic and time constraints, a complete physical examination cannot be performed. Therefore, we must be familiar enough with the examination to know which aspects are necessary and which can be omitted.

Clinical Examination

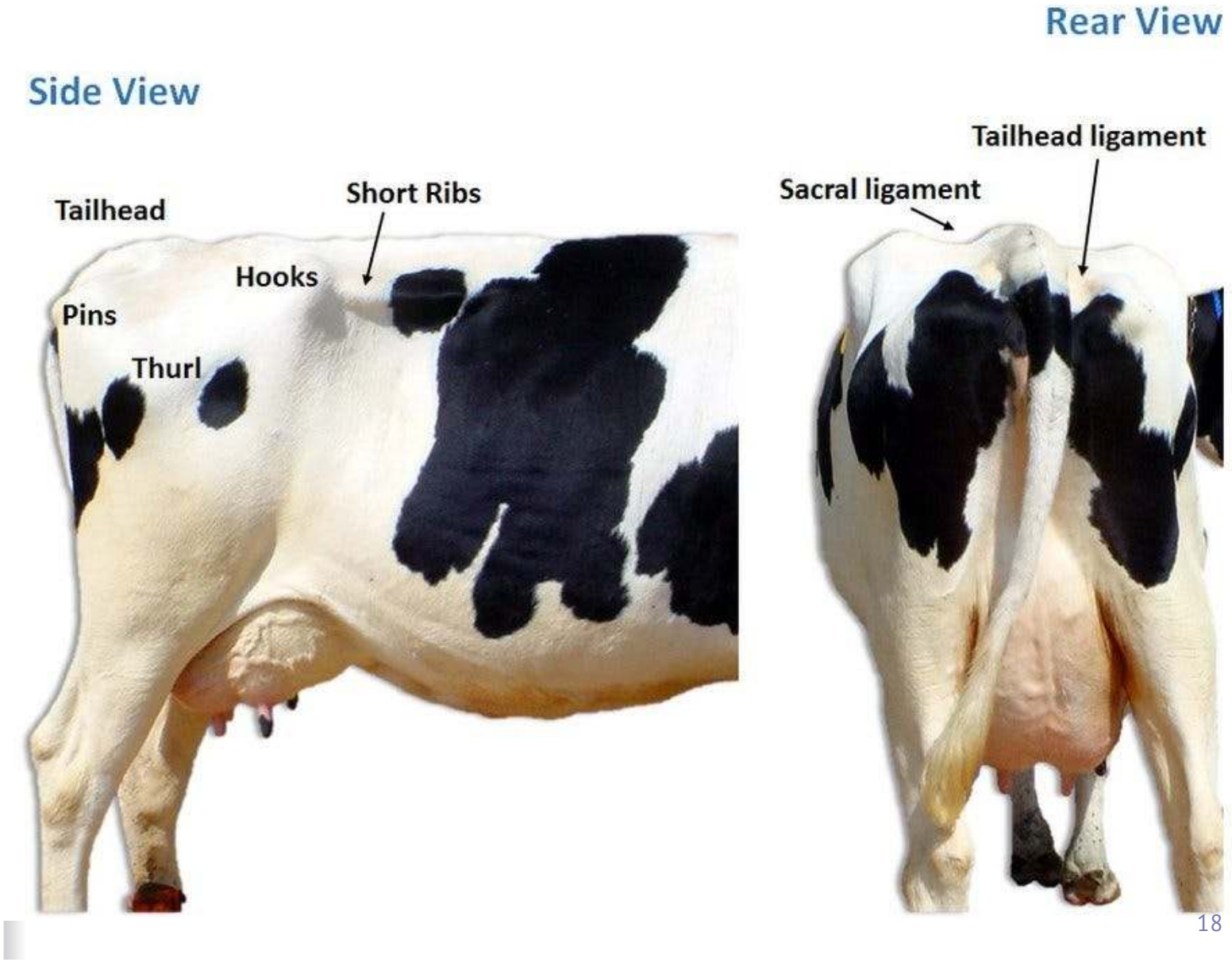
The first step in a physical examination is to observe the animal from a distance, then begin the examination from one side, move to the other side, examine the rear, and finally examine the head and neck.

Visual Examination

- posture
- gait
- behavior
- physical condition:
  - body condition score (BCS)
  - rumen fill (RF) score

Body Condition Score (BCS)

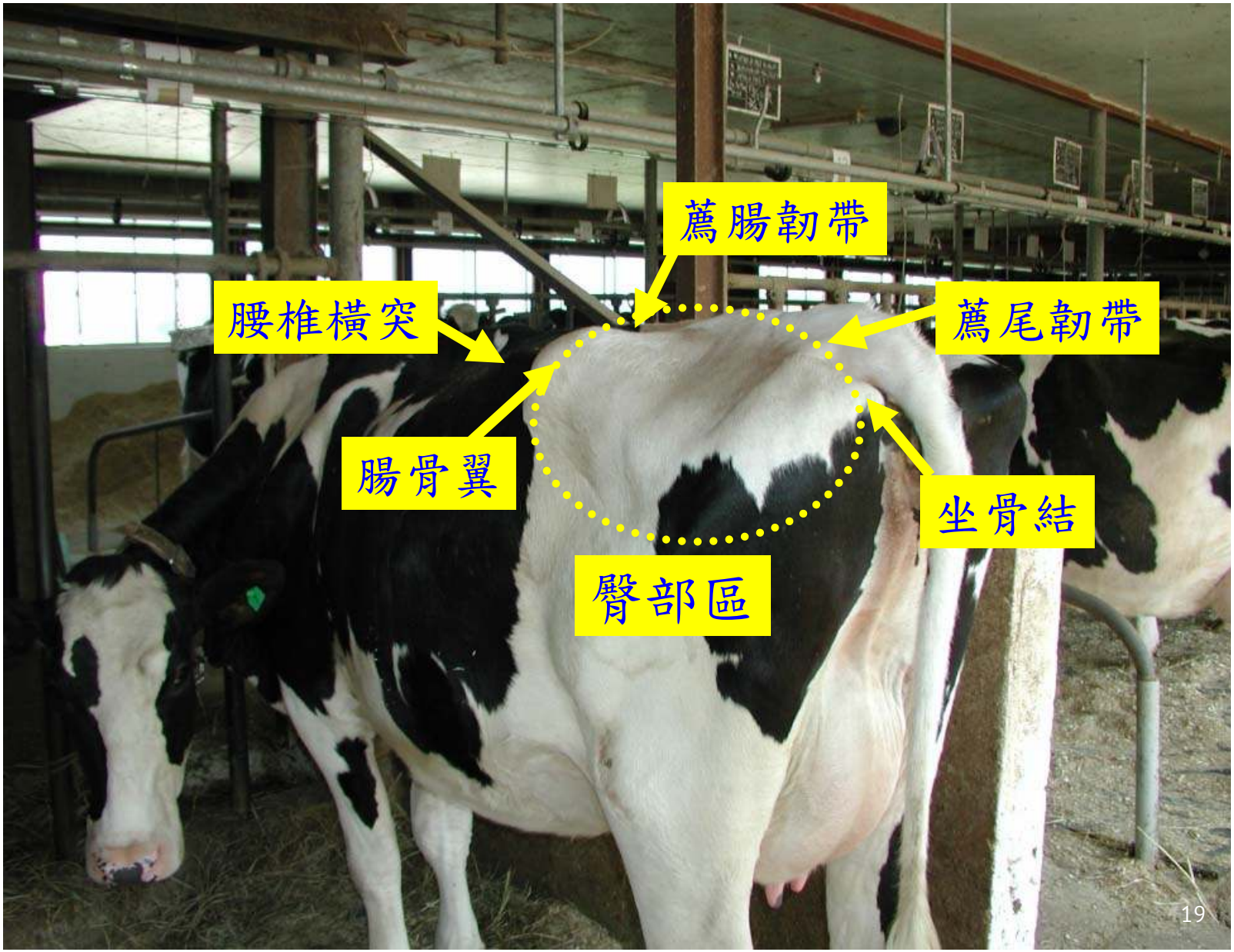
Common scoring system for dairy cow is a 1–5 scale, with increment of 0.25.

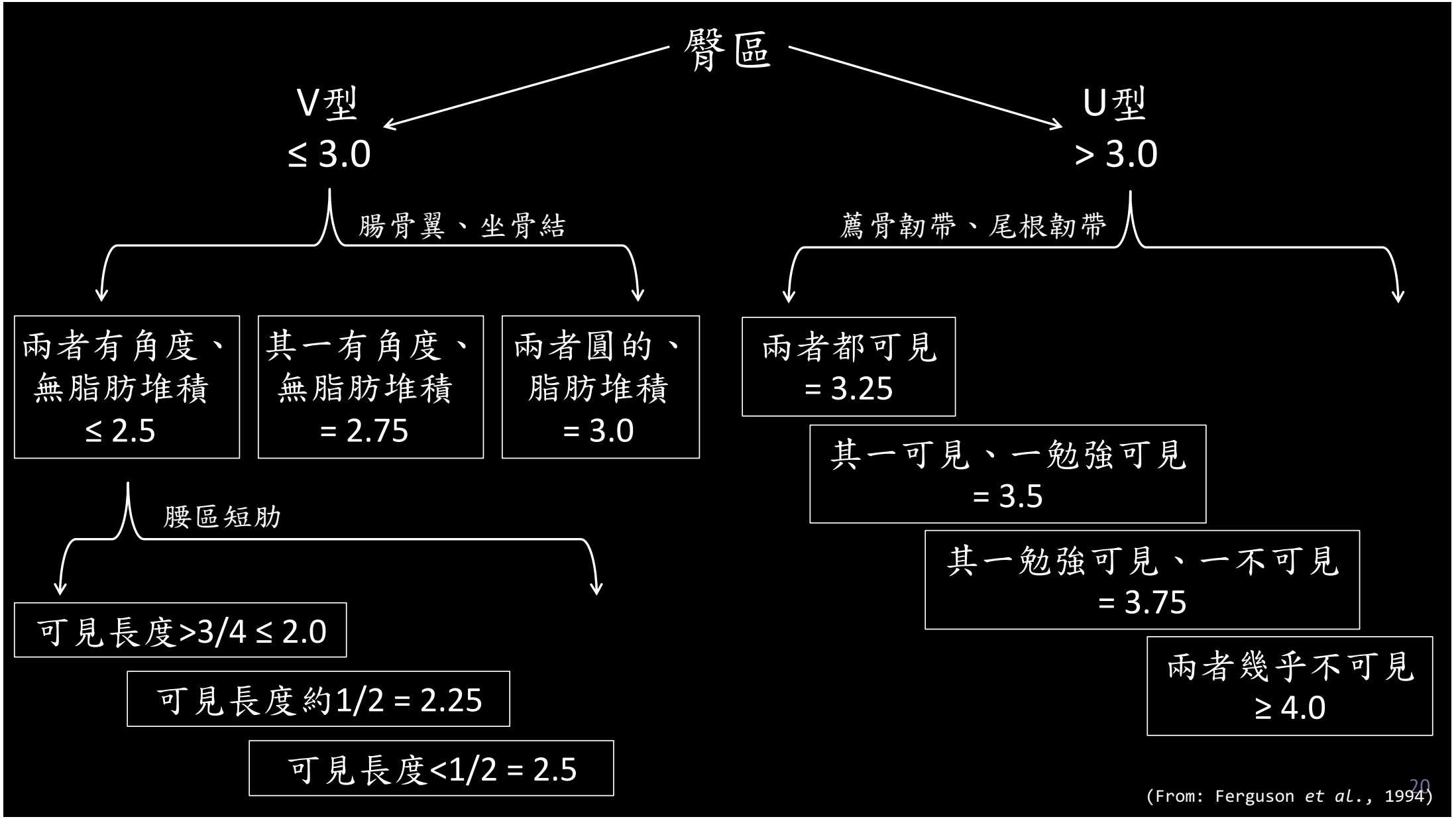


Body Condition Score (BCS)

Cow should be in a standing position.

Evaluation at a 45° angle form behind.





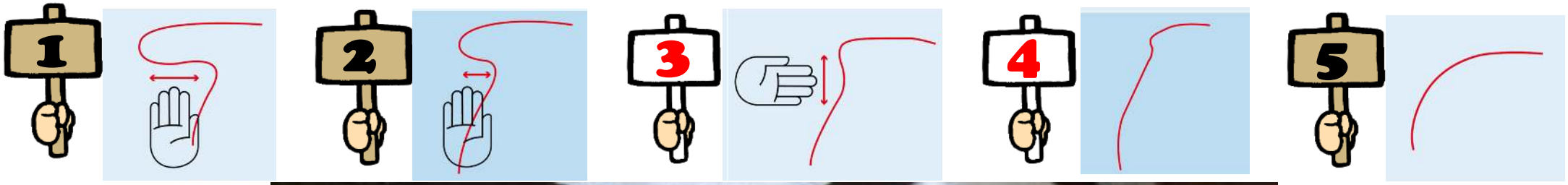
(From: Ferguson *et al.*, 1994)

Rumen Fill (RF) Score

A 1–5 scale

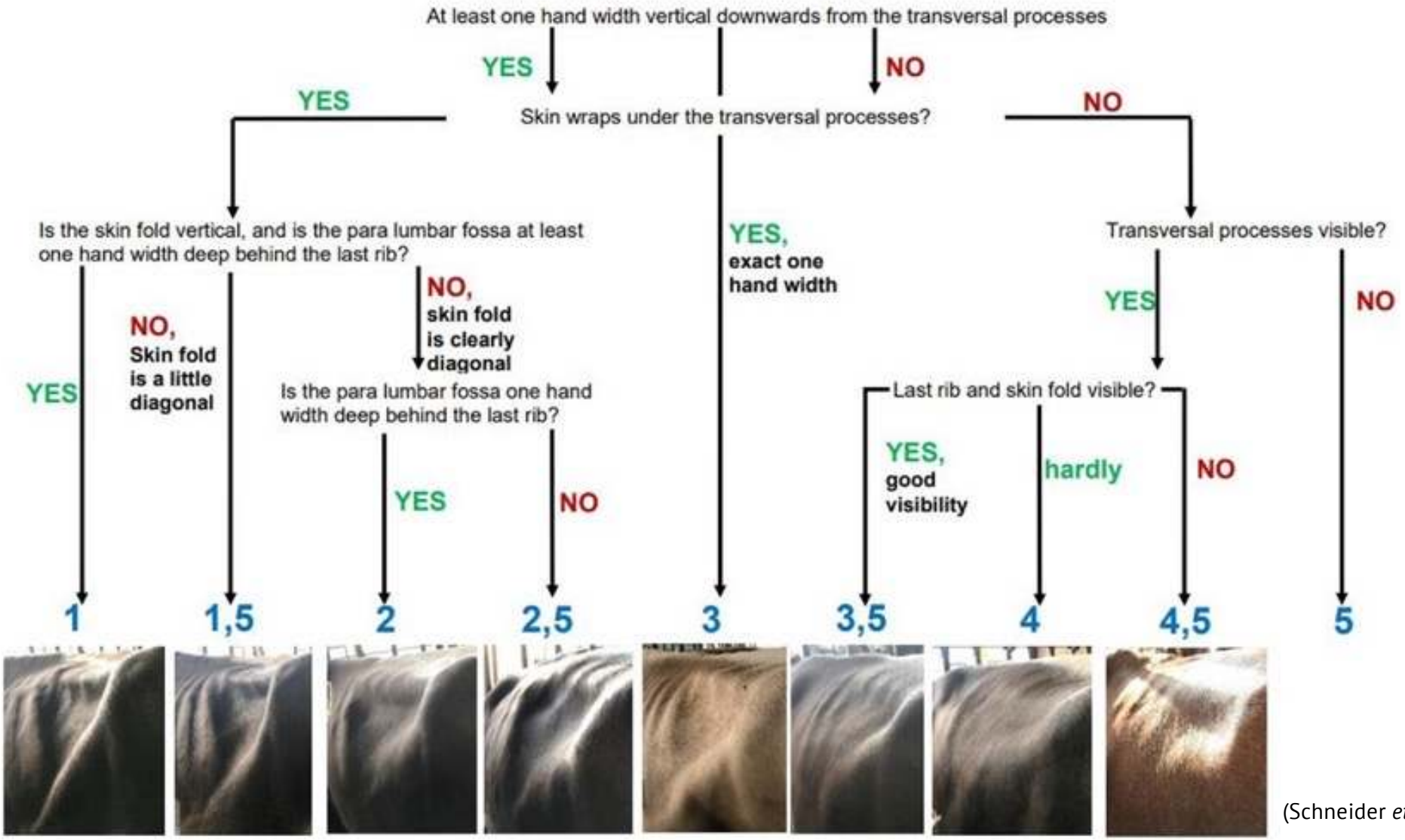


Rumen Fill (RF) Score





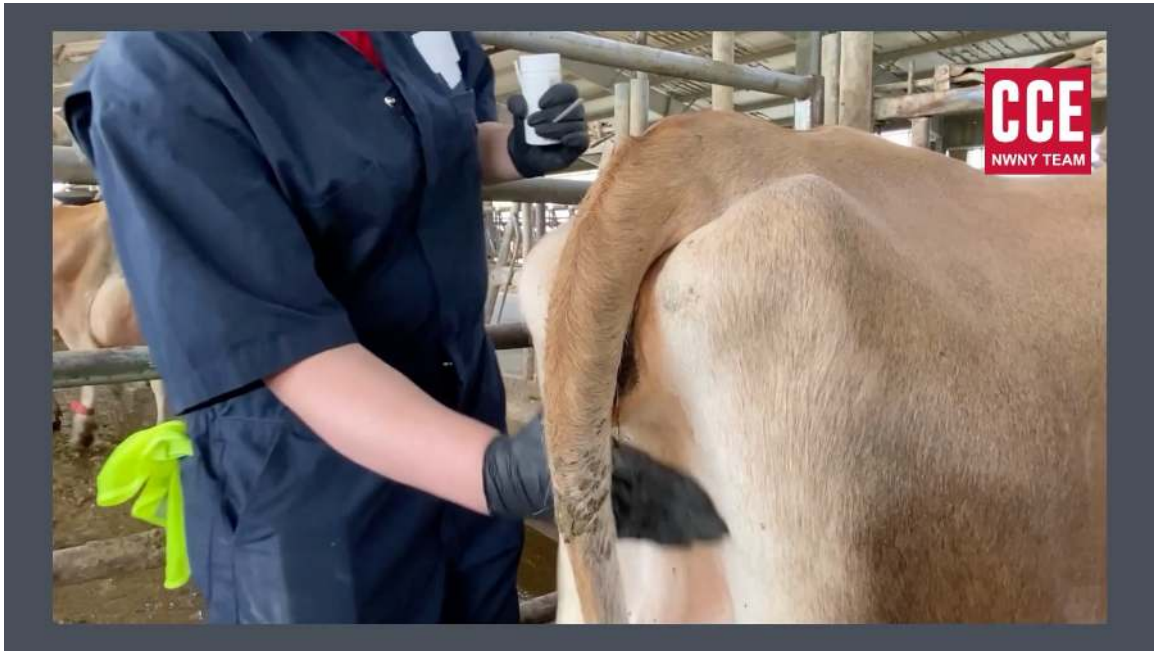
Scoring Rumen Fill



(Schneider *et al.*, 2022)

Physical Examination

- Obtaining a sample of urine





Physical Examination

- Body temperature
  - adult: 38.0–39.0 °C
  - calf: 38.5–39.5 °C
  - goat/sheep: 38.5–39.5 °C
  - kid/lamb: 39.0–40.0 °C



Physical Examination

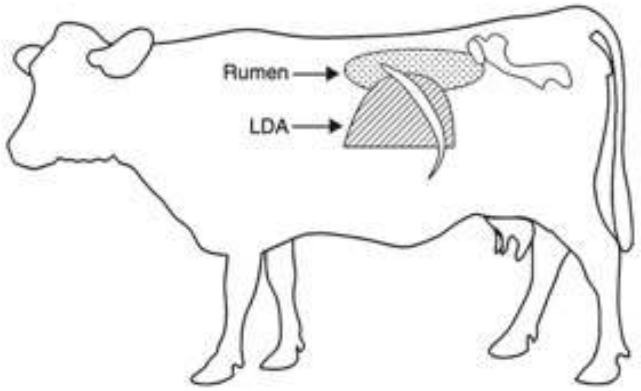
- Auscultation: Rumen
  - Rate: 3–5 times/2 min
  - Duration, strength

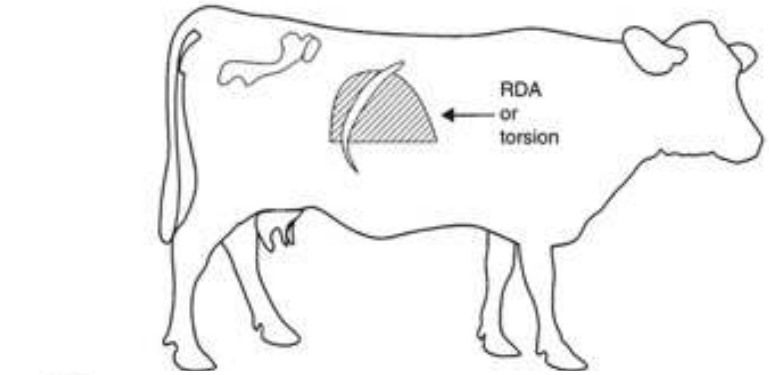


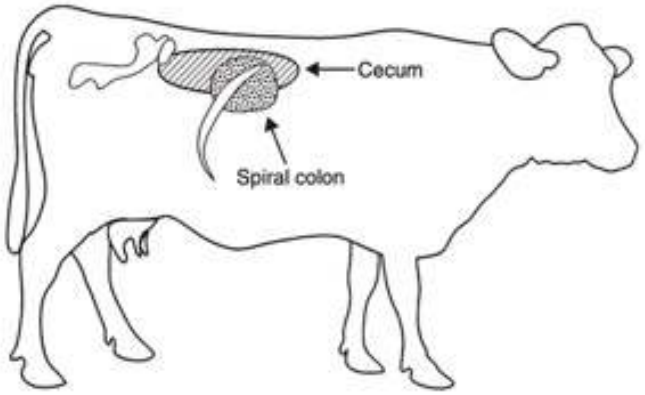
Physical Examination

- Auscultation with concurrent percussion









Physical Examination

- ballottement of the abdomen on the right side
  - "sloshing" sounds: intraabdominal fluid



Physical Examination

- Auscultation: Heart
  - Rate
    - adult: 60–80 bpm
    - calf: 70–90 bpm
    - goat/sheep: 70–110 bpm
    - kid/lamb: 110–140 bpm
  - Rhythm



Physical Examination

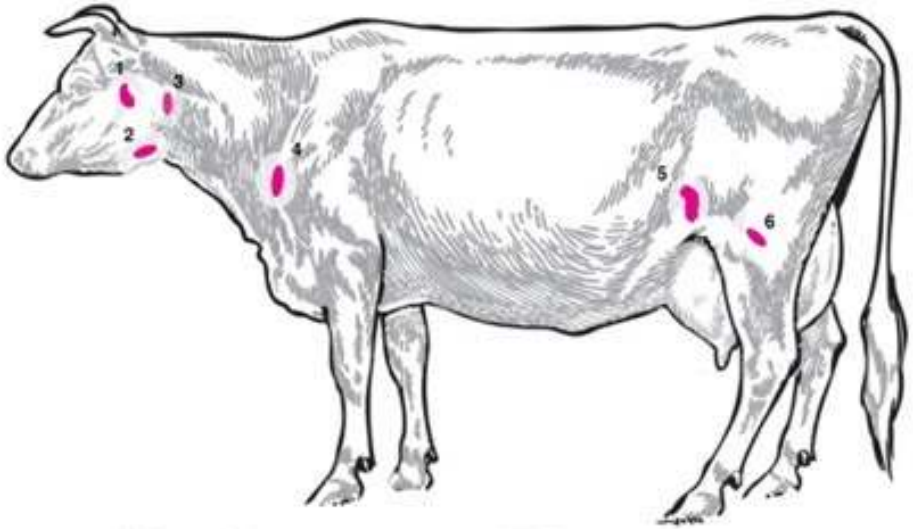
- Auscultation: respiratory
  - Rate
    - adult: 15–35 bpm
    - calf: 20–50 bpm
    - goat/sheep: 15–40 bpm
    - kid/lamb: 40–65 bpm
  - Lung sound





Physical Examination

- Palpation
  - Lymph node: parotid, submandibular, retropharyngeal, prescapular, supramammary
  - abdominal pain evidence
  - feet and leg



(1) Parotid
(2) Submandibular
(3) Retropharyngeal
(4) Prescapular
(5) Prefemoral
(6) Supramammary

(doi: 10.3791/57195)

Physical Examination

- Evaluate the head and neck regions
  - symmetric in appearance
  - discharges from the eyes, ears, nose, or mouth
  - oral cavity (mucosa color, lesions, foreign body)
  - hydration
- Rectal palpation

Ruminant Disease

Diseases of the Eye



Ophthalmic Examination

- Abnormal appearance eyes (asymmetry or color change)
- Presence of ocular discharge
- Presence of ocular pain
- Reduced vision or blindness
- Change in the performance or behavior of the patient
- Prepurchase examination
- Follow-up of a previously diagnosed ophthalmic condition
- Examinations for inherited ocular disorders

Sings of Ocular Disease

- Ocular/periocular asymmetry
- Change in globe position: exophthalmos, enophthalmos, strabismus
- Change in globe size: microphthalmia
- Change in eyelid conformation: entropion, ectropion, blepharitis, conjunctivitis, or ptosis
- Change in tissue volume: neoplasms
- Change in color
- Discharge
- Pain
- Blindness

Ocular Disease

- Trauma: eye, lids, orbit
- Infectious ocular diseases:
  - Mycoplasma infection
  -
- Neoplasia:

Endophthalmitis



https://reurl.cc/rk00k4

- Inflammation of the interior eyeball:
  - endogenous: sepsis
  - exogenous: trauma, infectious bovine keratoconjunctivitis (IBK)
- Signs:
  - eye opacity, visual impairment, enlarged or shrinking eyeball
- Diagnosis: ocular ultrasound (retinal detachment)
- Treatment:
  - systemic antibiotics, NSAIDs, topical antibiotic ointment, enucleation

Panophthalmitis

- Inflammation of the eyeball and the surroundings:
  - IBK, malignant catarrhal fever (vascular damage), tuberculosis (granulomatous lesion), trauma
- Signs:
  - exophthalmos, eyelid edema, ocular opacity, pus, secondary orbital cellulitis, systemic illness
- Treatment:
  - enucleation, systemic antibiotics, NSAIDs



https://reurl.cc/OQ66Dr

Orbital Cellulitis/Inflammation

- Cause:
  - Trauma of the eyelids or conjunctiva, further infected by bacteria
  - Endophthalmitis → panophthalmatis
- Signs:
  - Acute: warm, painful swelling of the orbital region, eyelids, and nictitans, conjunctival edema, and a degree of exophthalmos
  - Chronic: localized soft-tissue



https://veteriankey.com/ocular-diseases/

Orbital Cellulitis/Inflammation

- Diagnosis:
  - CBC: normal or mild neutrophilia
  - Differential diagnosis: neoplasia, chronic frontal sinusitis
- Treatment:
  - acute: systemic broad-spectrum antibiotics, NSAID, topical antibiotic ointment or lubricant, warm compresses of the orbit (tid, 5 min/time)
  - chronic: surgical drainage and above included

Inflammation of the eye

- *Mycoplasma* infection in goats and sheep:
  - *M. conjunctivae*, *M. mycoides*, *M. agalactiae*, *M. arginini*, *Acholeplasma oculusi*, etc
  - Transmission: direct contact, vectors
  - other clinical signs: arthritis, mastitis
- *Chlamydia* infection in goats and sheep:
  - *Chlamydia pecorum*
  - Transmission: direct contact
  - other clinical signs: abortion

Inflammation of the eye

- Infectious bovine rhinotracheitis (bovine herpesvirus type 1):
- Malignant catarrhal fever
- Bovine viral diarrhea
- Bluetongue
- Bovine leukemia virus
- Peste des petits ruminants
- Scrapie

Inflammation of the eye

- Infectious bovine keratoconjunctivitis (IBK)/pink eye:
  - most common ocular disease of cattle (especially in calves), affecting one eye
  - *Moraxella bovis*
  - corneal ulceration/edema, photophobia, blepharospasm, and lacrimation
  - controlling face fly (*Musca autumnalis*)



Inflammation of the eye

- Infectious bovine keratoconjunctivitis (*Moraxella bovis*)
- *Listeria monocytogenes*
- *Mycoplasma* spp.
- Tuberculosis (*Mycobacterium bovis*)
- Infectious thromboembolic meningoencephalitis (*Histophilus somni*)
- Leptospirosis

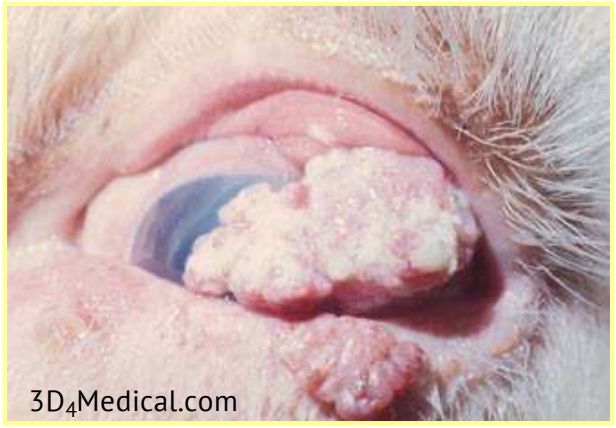
Ocular Neoplasia

Although there are many types of primary and secondary tumors affecting the eyes and surroundings, tissue deformity and functional loss are the most common complaints. Treatment strategies for ocular tumors in livestock can be various, from curative or alleviate discomfort (enucleation) to eradicating the problem (culling).

The classification of tumors, “benign vs. malignant" or "limited vs. systemic," can be used as one of the criteria for selecting a treatment for ocular neoplasia.

Ocular Squamous Cell Carcinoma (OSCC)

- The most economically important neoplasm of cattle
  - carcass condemnations, production losses, treatment costs.
- The most common tumor affecting cattle in North America (≈12.5%)
- Location: limbus, nictitating membrane and eyelids, medial canthal regions



OSCC







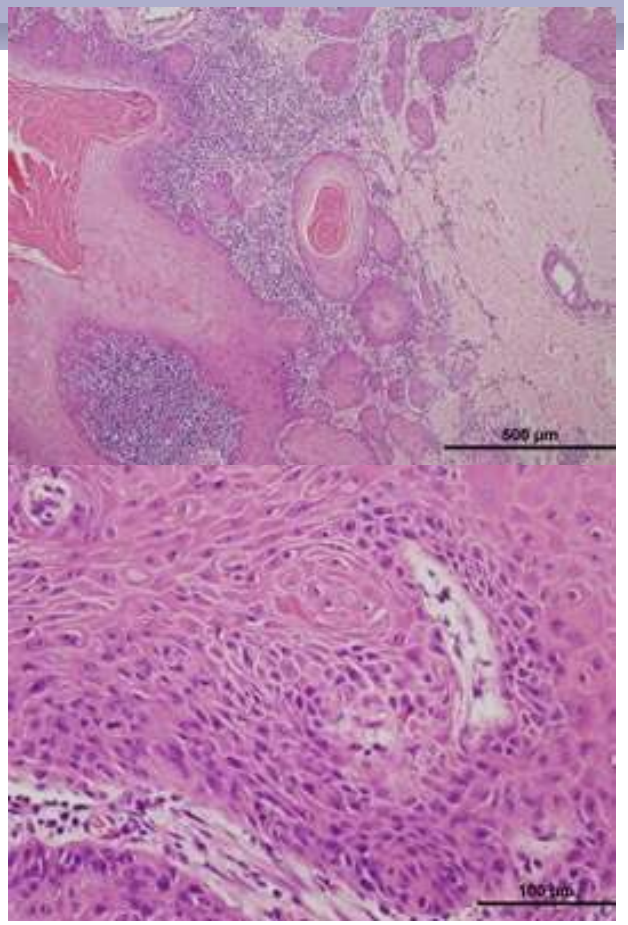
- Risk factors:
  - UV, non-pigmentation, age, bread
- Stage of malignancy:
  - Premalignant SCC: small, white, superficial, and elevated hyperplastic plaques or papilloma-like structures
  - Malignant SCC: more irregular, nodular, pink, erosive, and invasive

OSCC

- Invasion/metastasis :
  - Nictitating membrane tumors can easily invade to the base of the membrane and cartilage.
  - Eyeball, conjunctiva, orbit, bones of the orbit, sinuses, and regional lymph nodes
  - (rare) Thoracic and abdominal organ

OSCC

- Pathology:
  - Benign: superficial anucleated, keratinized squamous cells and deeper epithelial cells with enlarged nuclei and coarse chromatin clumping; basement membrane not invaded
  - Malignant: pleomorphic cells, large hyperchromatic nuclei, and prominent nucleoli; invasion across the basement membrane is noted, and keratin-pearl formation or intercellular bridges is usually present



OSCC

- Treatment:
  - Surgical excision: local, enucleation, exenteration
  - Cryotherapy (lesion < 0.5 cm)
  - Radiofrequency hyperthermia
- Recurrence rates range between 30.4% and 42.4%

Lymphosarcoma

- Systemic neoplastic disease of the lymphoreticular tissue
  - bovine leukemia virus (BLV) infection
  - most common cause of orbital neoplasia in cattle
- Clinical signs: exophthalmos caused by orbital invasion of neoplastic lymphoid cells



https://reurl.cc/dOpbWq

Lymphosarcoma

- Treatment: Enucleation
- Most cattle with orbital lymphosarcoma either die or are euthanized in terminal stages of disease within 6 months of diagnosis
- Control of bovine lymphosarcoma depends on efforts to eradicate bovine leukemia virus (blood-borne disease)


Thank you.